

Micro Energy Harvesting 2026

Exercise 2: Photovoltaics

Task 1: Irradiance

On a cloudless summer day ($I_D = 850 \text{ W/m}^2$) the sun is at an angle of 45° towards the horizon.

- a) What is the optimal orientation of the solar module and what is the irradiance?
- b) What is the irradiance at an angle of the solar module of 15° towards the horizon?

Now a winter day is considered: $I_D = I_{Diff} = 300 \text{ W/m}^2$. The sun is at an angle of 23° towards the horizon. What is the irradiance for a solar module under the two angles of attack used above?

Task 2: Silicon solar cell

A silicon solar cell (band gap = 1.12 eV) with negligible parasitic resistance is uniformly illuminated by powerful sodium vapor lamps (total 1 kW/m^2 at quasi-monochromatic 589 nm). The square cell has an edge length of 5 cm and an ideality factor of 1.1. The dark current I_0 is 1 pA at 300 K.

- a) What is the short-circuit current I_{SC} , open-circuit voltage V_{OC} , fill factor FF and efficiency η of the solar cell (assuming ideal QE and 100 % collection of all e^-h^+ pairs).
- b) Corrosion on the front metal contacts leads to a series resistance R_S of 0.01Ω . How high is the proportion of power loss due to series resistance compared to the original cell?

Task 3: Indoor Photovoltaics

A solar cell from Solems (type 07/055/020; www.solems.com) is to be used in a solar energy harvesting system.

- a) Determine and compare the FF and cell efficiency for the different lighting scenarios.
- b) Which battery system would you combine with this solar cell and how?
- c) What else can the solar cell be used for besides supplying energy to the system?

Bonus Task 4: Position of the sun

On a clear November 12th, at 10:00 am local time in Toronto (Canada), someone wants to know the a) Hour Angle (HRA), b) Elevation Angle α and the Azimuth of the Sun.